

Multiplying Binomials

Name
Date
Course

Expand.

$$\textcircled{1} (2x+3)(4x+7) = 8x^2 + 14x + 12x + 21 = \boxed{8x^2 + 26x + 21}$$

$$\textcircled{2} (6y+4)(8y-3) = 48y^2 - 18y + 32y - 12 = \boxed{48y^2 + 14y - 12}$$

$$\textcircled{3} (z-5)(3z+2) = 3z^2 + 2z - 15z - 10 = \boxed{3z^2 - 13z - 10}$$

$$\textcircled{4} (2x-7)(2x-5) = 4x^2 - 10x - 14x + 35 = \boxed{4x^2 - 24x + 35}$$

$$\textcircled{5} (y+2)(y+4) = y^2 + 4y + 2y + 8 = \boxed{y^2 + 6y + 8}$$

$$\textcircled{6} (6z+7)(z-1) = 6z^2 - 6z + 7z - 7 = \boxed{6z^2 + z - 7}$$

$$\textcircled{7} (2x+3)(9x+4) = 18x^2 + 8x + 27x + 12 = \boxed{18x^2 + 35x + 12}$$

$$\textcircled{8} (8y-1)(2y-1) = 16y^2 - 8y - 2y + 1 = \boxed{16y^2 - 10y + 1}$$

$$\textcircled{9} (z-6)(3z+4) = 3z^2 + 4z - 18z - 24 = \boxed{3z^2 - 14z - 24}$$

$$\textcircled{10} (10x-7)(3x-2) = 30x^2 - 20x - 21x + 14 = \boxed{30x^2 - 41x + 14}$$

$$\textcircled{11} (2y-5)(3y+5) = 6y^2 + 10y - 15y - 25 = \boxed{6y^2 - 5y - 25}$$

$$\textcircled{12} (-7z+1)(2z-4) = -14z^2 + 28z + 2z - 4 = \boxed{-14z^2 + 30z - 4}$$

$$\textcircled{13} (-2x+3)(-3x+4) = 6x^2 - 8x - 9x + 12 = \boxed{6x^2 - 17x + 12}$$

$$\textcircled{14} (4x^2-9)(2x+3) = \boxed{8x^3 + 12x^2 - 18x - 27}$$

$$\textcircled{15} (-3y^5+2)(4y^2-1) = \boxed{-12y^7 + 3y^5 + 8y^2 - 2}$$

$$\textcircled{16} (2y+5)^2 = (2y+5)(2y+5) = 4y^2 + 10y + 10y + 25 = \boxed{4y^2 + 20y + 25}$$

$$\textcircled{17} (6z-5)^2 = (6z-5)(6z-5) = 36z^2 - 30z - 30z + 25 = \boxed{36z^2 - 60z + 25}$$

$$\textcircled{18} (9x+1)^2 = (9x+1)(9x+1) = 81x^2 + 9x + 9x + 1 = \boxed{81x^2 + 18x + 1}$$

$$\textcircled{19} (5x-2)^2 = (5x-2)(5x-2) = 25x^2 - 10x - 10x + 4 = \boxed{25x^2 - 20x + 4}$$

$$\textcircled{20} (-x+3)^2 = (-x+3)(-x+3) = x^2 - 3x - 3x + 9 = \boxed{x^2 - 6x + 9}$$

$$\textcircled{21} (2a+5b)(3a+4b) = 6a^2 + 8ab + 15ab + 20b^2 = \boxed{6a^2 + 23ab + 20b^2}$$

$$\textcircled{22} (g+5h)(6g-4h) = 6g^2 - 4gh + 30gh - 20h^2 = \boxed{6g^2 + 26gh - 20h^2}$$

$$\textcircled{23} (3x+2y)(x-y) = 3x^2 - 3xy + 2xy - 2y^2 = \boxed{3x^2 - xy - 2y^2}$$

$$\textcircled{24} (2r-5t)^2 = (2r-5t)(2r-5t) = 4r^2 - 10rt - 10rt + 25t^2 = \boxed{4r^2 - 20rt + 25t^2}$$

$$\textcircled{25} (x-3y)^2 = (x-3y)(x-3y) = x^2 - 3yx - 3yx + 9y^2 = \boxed{x^2 - 6yx + 9y^2}$$

$$\begin{aligned} \textcircled{26} (5x+1)(2x^2-6x+3) &= 10x^3 - 30x^2 + 15x + 2x^2 - 6x + 3 \\ &= \boxed{10x^3 - 28x^2 + 9x + 3} \end{aligned}$$

$$\begin{aligned} \textcircled{27} (5y^2+6y-2)(y+3) &= 5y^3 + 6y^2 - 2y + 15y^2 + 18y - 6 \\ &= \boxed{5y^3 + 21y^2 + 16y - 6} \end{aligned}$$

$$\begin{aligned} \textcircled{28} (z-2)(3z^2+6z-8) &= 3z^3 + 6z^2 - 8z - 6z^2 - 12z + 16 \\ &= \boxed{3z^3 - 20z + 16} \end{aligned}$$

$$\begin{aligned} \textcircled{29} (2x^2+x-1)(4x-3) &= 8x^3 + 4x^2 - 4x - 6x^2 - 3x + 3 \\ &= \boxed{8x^3 - 2x^2 - 7x + 3} \end{aligned}$$

$$\begin{aligned}\textcircled{30} (2y-5)(3y^2-2y+5) &= 6y^3-4y^2+10y-15y^2+10y-25 \\ &= \boxed{6y^3-19y^2+20y-25}\end{aligned}$$

$$\begin{aligned}\textcircled{31} (4z^2-2z+5)(6z-5) &= 24z^3-12z^2+30z-20z^2+10z-25 \\ &= \boxed{24z^3-32z^2+40z-25}\end{aligned}$$

$$\textcircled{32} (-5x+7)(2x^2-4x-2)$$

$$\textcircled{33} (-2x+5)(-2x^2+5x+1)$$